

CS & IT ENGINEERING



Digital Logic

Optimisation

Lecture No. 04



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Recap of Previous Lecture



Topic



Imp Concepts
+
Problem Solving

Topics to be Covered

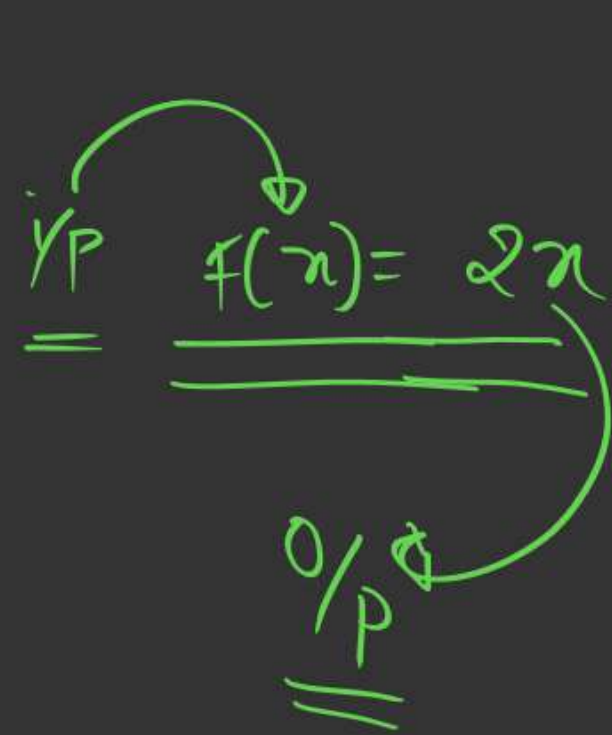


Topic



- Min-terms,
- Max-terms,
- Intro to k-map.

Boolean Function:



↳ a function of i/p's on which the output depends.


$$F(P, Q) = P\bar{Q} + \bar{Q}P$$

1) no. of variables $\Rightarrow$ (2)

2) no. of terms $\Rightarrow$ (2)

Boolean Function

Sum OF Products
(SOP)

Product of Sums
(POS)

→ Minterms

→ Maxterms

→ 1s

→ 0s

→ $A = 1$
 $\bar{A} = 0$

→ $\bar{A} = 1$

→ $A = 0$

Minterm = Maxterm

Standard Canonical Form



eg1: $F(P, Q) = P + Q$ $\rightarrow$ not a Standard Canonical Form (SCF)

eg2: $F(P, Q) = PQ + \overline{P}\overline{Q}$ $\rightarrow$ valid SCF

n ip

$Q^n \rightarrow \min$

$Q^n \rightarrow \max$

$(x \cdot y)$ $(x+y)$
 $\uparrow$ $\uparrow$
SOP POS

eg Random eg

Decimel	QOR	Minterm	Maxterm	$f(P, Q, R)$
0	000	$\overline{P} \cdot \overline{Q} \cdot \overline{R}$	$P+Q+R$	1
1	001	$\overline{P} \cdot \overline{Q} \cdot R$	$P+Q+\overline{R}$	0
2	010	$\overline{P} \cdot Q \cdot \overline{R}$	$P+\overline{Q}+R$	1
3	011	$\overline{P} \cdot Q \cdot R$	$P+\overline{Q}+\overline{R}$	1
4	100	$P \cdot \overline{Q} \cdot \overline{R}$	$\overline{P}+Q+R$	0
5	101	$P \cdot \overline{Q} \cdot R$	$\overline{P}+Q+\overline{R}$	0
6	110	$P \cdot Q \cdot \overline{R}$	$\overline{P}+\overline{Q}+R$	0
7	111	$P \cdot Q \cdot R$	$\overline{P}+\overline{Q}+\overline{R}$	1

$$\begin{aligned} & \overline{\overline{P} \cdot \overline{Q} \cdot \overline{R}} \\ &= \overline{\overline{P}} + \overline{\overline{Q}} + \overline{\overline{R}} \\ &= P + Q + R \end{aligned}$$

① Standard Canonical SOP form:-

$$F(P, Q, R) = \overline{P}\overline{Q}\overline{R} + \overline{P}Q\overline{R} + \overline{P}QR + PQR$$

000	010	011	111
↓	↓	↓	↓
0	2	3	7

minterms

(m_i)

$$= m_0 + m_2 + m_3 + m_7$$

$$= \sum m(0, 2, 3, 7)$$

$$= \Sigma(0, 2, 3, 7)$$

② Standard Canonical POS Form:-

Maxterms

↓
(M_i)

$$F(P, Q, R) = (P + Q + \bar{R})(\bar{P} + Q + R)(\bar{P} + Q + \bar{R})(\bar{P} + \bar{Q} + R)$$

001

↓
1

100

↓
4

101

↓
5

110

↓
6

$$= M_1 \cdot M_4 \cdot M_5 \cdot M_6$$

$$= \Pi M(1, 4, 5, 6)$$

$$= \Pi(1, 4, 5, 6)$$

(Q) What are the no. of terms present in the Standard Canonical SOP form
37.5%

$$F(P, Q, R) = QR + P$$

Soln:-

variables: ³
 (P, Q, R)

$$\begin{aligned} &= Q \cdot R \cdot 1 + P \cdot 1 \cdot 1 \\ &= Q \cdot R \cdot (P + \bar{P}) + P \cdot (Q + \bar{Q}) \cdot (R + \bar{R}) \\ &= Q \cdot R \cdot P + Q \cdot R \cdot \bar{P} + (PQ + P\bar{Q})(R + \bar{R}) \end{aligned}$$

$$= \cancel{PQR} + \bar{P}QR + PQR + PQ\bar{R} + \bar{P}QR + P\bar{Q}\bar{R}$$

$$= \bar{P}QR + PQR + PQ\bar{R} + P\bar{Q}R + P\bar{Q}\bar{R}$$

Ans. $\rightarrow$ (5)

★ Consensus Thm $\implies$ Redundancy Thm.

$$F(P, Q, R) = PQ + \bar{P}R + \underbrace{(QR)}_{\text{Red}} = PQ + \bar{P}R$$

Ex:-

$$\begin{aligned} &= PQ + \bar{P}R + (P + \bar{P})QR \\ &= PQ + \bar{P}R + PQR + \bar{P}QR \\ &= PQ(1+R) + \bar{P}R(1+Q) \\ &= PQ + \bar{P}R \end{aligned}$$

K-MAP

k-map

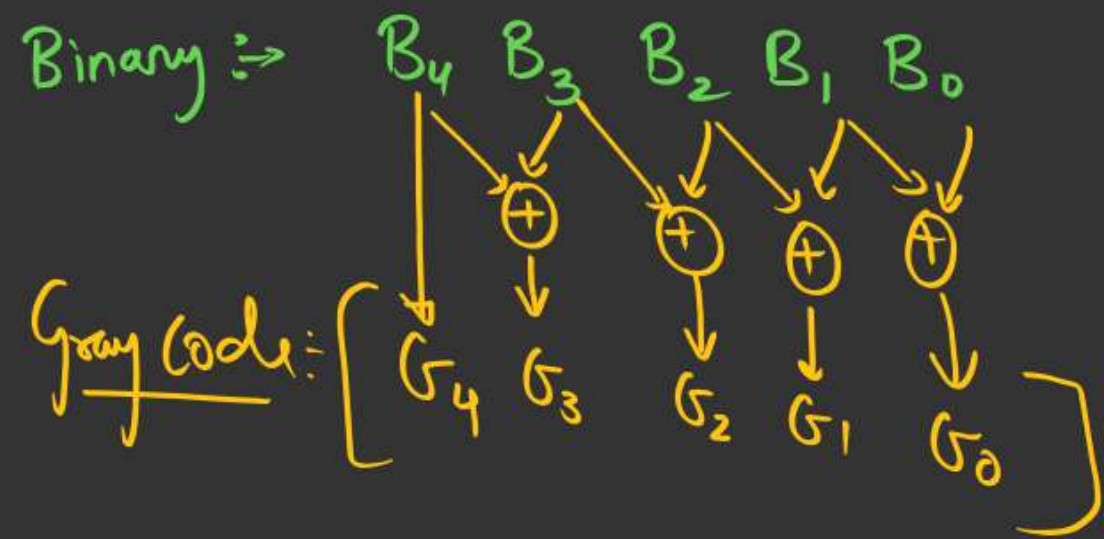
→ also known as
Modified Veitch Diagram.

→ Karnaugh Map

Minimization by k-map

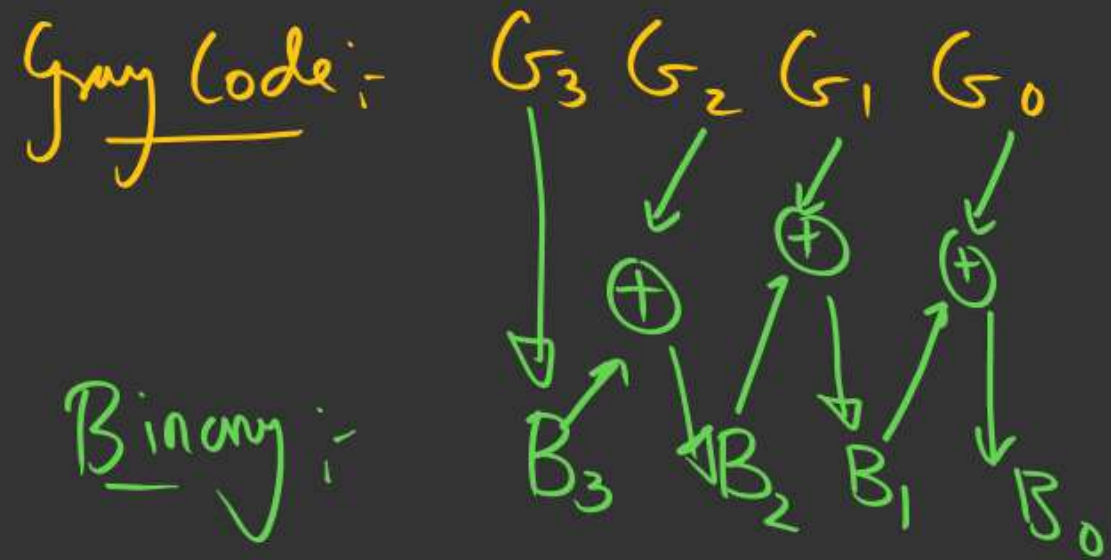
→ uses Gray Code
(based on)

Binary $\longrightarrow$ Gray Code



eg- Binary: 0110101
 $\downarrow$
Gray Code: 0101111

Gray Code $\longrightarrow$ Binary



Gray: 1101011

Binary: 1001101 ✓

Gray



Binary

10110001



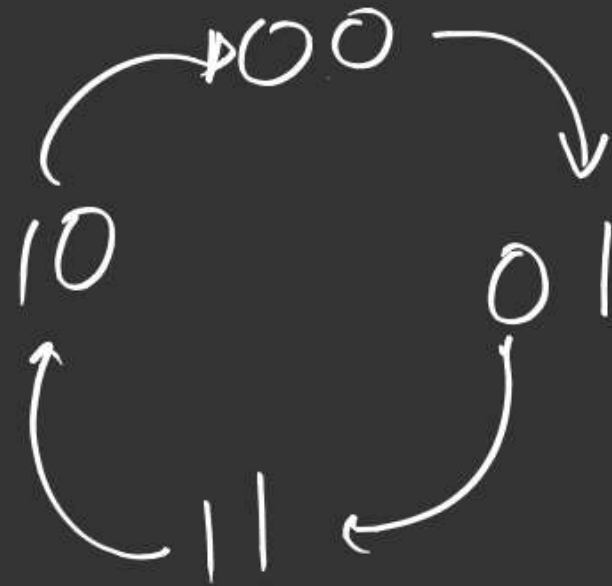
11011110



2 variable

Decimal	Binary	Gray
0	00	00
1	01	01
2	10	11
3	11	10

Successive/consecutive numbers differ by 1 bit.

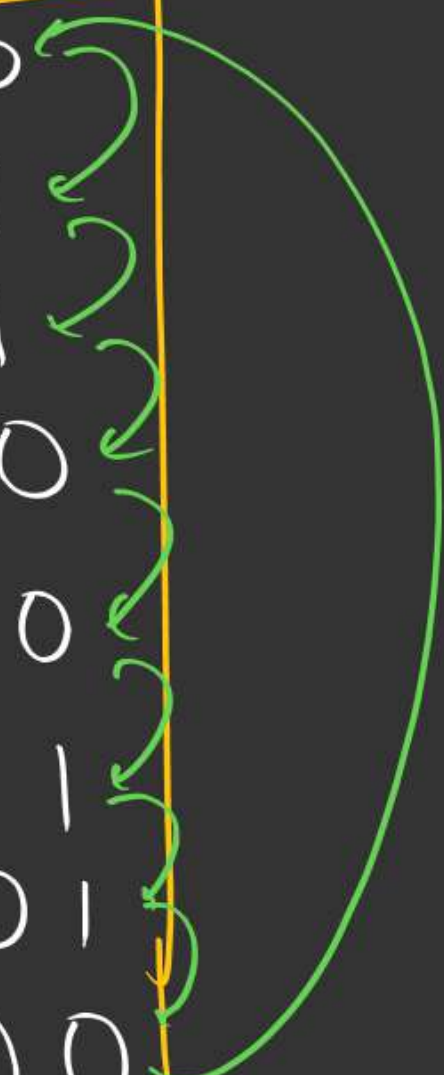


Gray Code :-

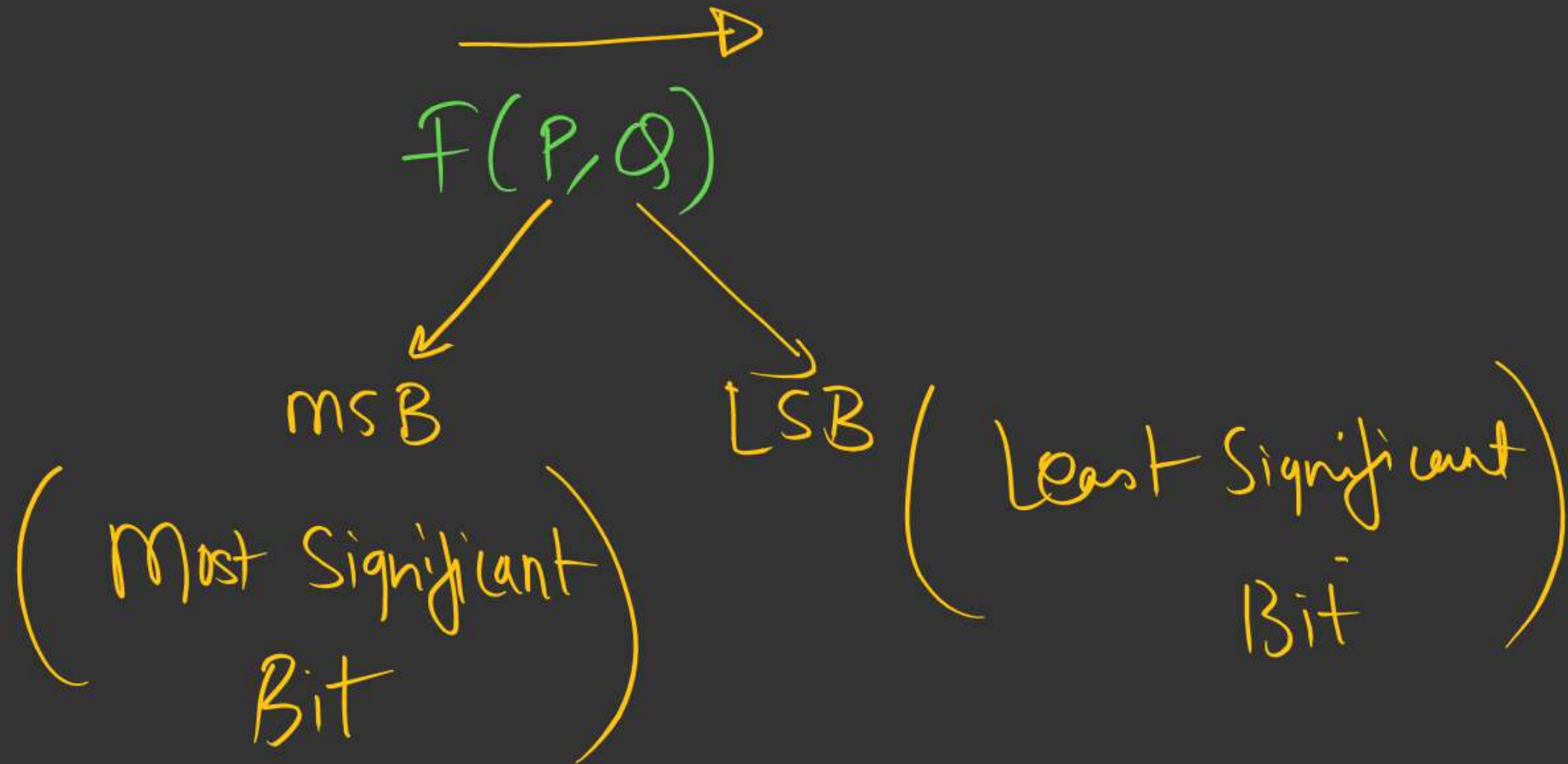
- 1) Unity Hamming distance Code
- 2) Cyclic code
- * 3) Reflecting code

3 variables

Decimal	Binary	Gray
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100



(703)



2 variable:

$F(A,B)$

	$\overline{B}$	B
$\overline{A}$	$\overline{A} \cdot \overline{B}$ 00	$\overline{A} \cdot B$ 01
A	$A \cdot \overline{B}$ 10	$A \cdot B$ 11

3 variables

$f(A, B, C)$

		BC	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
A		→ 00	01	11	10	
$\overline{A}$	0	000 0	001 1	011 3	010 2	
A	1	100 4	101 5	111 7	110 6	

4 variables:- $F(A, B, C, D)$

		CD	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$
		00	01	11	10
$\bar{A}\bar{B}$	00	0000 0	0001 1	0011 3	0010 2
$\bar{A}B$	01	0100 4	0101 5	0111 7	0110 6
AB	11	1100 12	1101 13	1111 15	1110 14
$A\bar{B}$	10	1000 8	1001 9	1011 11	1010 10



2 mins Summary



Min term

Max term

K map



THANK - YOU

